

**Mathematical Programming and
Operations Research**
Modeling, Algorithms, and Complexity
Examples in Excel and Python
(Work in progress)

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Preface

This entire book is a working manuscript. The first draft of the book is yet to be completed.

This book is being written and compiled using a number of open source materials. We will strive to properly cite all resources used and give references on where to find these resources. Although the material used in this book comes from a variety of licences, everything used here will be CC-BY-SA 4.0 compatible, and hence, the entire book will fall under a CC-BY-SA 4.0 license.

MAJOR ACKNOWLEDGEMENTS

I would like to acknowledge that substantial parts of this book were borrowed under a CC-BY-SA license. These substantial pieces include:

- "A First Course in Linear Algebra" by Lyryx Learning (based on original text by Ken Kuttler). A majority of their formatting was used along with selected sections that make up the appendix sections on linear algebra. We are extremely grateful to Lyryx for sharing their files with us. They do an amazing job compiling their books and the templates and formatting that we have borrowed here clearly took a lot of work to set up. Thank you for sharing all of this material to make structuring and forming this book much easier! See subsequent page for list of contributors.
- "Foundations of Applied Mathematics" with many contributors. See <https://github.com/Foundations-of-Applied-Mathematics>. Several sections from these notes were used along with some formatting. Some of this content has been edited or rearranged to suit the needs of this book. This content comes with some great references to code and nice formatting to present code within the book. See subsequent page with list of contributors.
- "Linear Inequalities and Linear Programming" by Kevin Cheung. See <https://github.com/dataopt/lineqlpbook>. These notes are posted on GitHub in a ".Rmd" format for nice reading online. This content was converted to $\text{\LaTeX}$ using Pandoc. These notes make up a substantial section of the Linear Programming part of this book.
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Part I

Linear Programming

1. Modeling: Linear Programming

Outcomes

1. Define what a linear program is
2. Understand how to model a linear program
3. View many examples and get a sense of what types of problems can be modeled as linear programs.

Resources

- Youtube! - Modeling with Linear Programming - Watch 0:00 - 4:19
- Youtube! - Watch 0:00 - 1:43

Linear Programming, also known as Linear Optimization, is the starting point for most forms of optimization. It is the problem of optimization a linear function over linear constraints.

In this chapter, we will define what this means, how to setup a linear program, and discuss many examples. Examples will be connected with code in Excel and Python (using with PuLP or Gurobipy modeling tools) so that you can easily start solving optimization problems. Tutorials on these tools will come in later chapters.

As a warm up, consider the following word problem.

Question: Find the age of Justin and Brad, given that the sum of their ages is equal to 76 and after 7 years, the age of Brad will be twice the age of Justin.

Solution: Let x and y denote the age of Justin and Brad, respectively. Then

$$x + y = 76 \tag{1}$$

$$2(x + 7) = y + 7. \tag{2}$$

On solving the foregoing equations we get

$$x = 23 \quad \text{and} \quad y = 53.$$

We will need to think about assigning variables to decisions and instead of equations guiding these variables, we may have inequalities that bound these variables in different ways. Thus, there can be many possible solutions.

1.1 Introductory LP Example

We begin with a simple example.

Example 1.1: Toy Maker

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Consider the problem of a toy company that produces toy planes and toy boats. The toy company can sell its planes for \$10 and its boats for \$8 dollars. It costs \$3 in raw materials to make a plane and \$2 in raw materials to make a boat. A plane requires 3 hours to make and 1 hour to finish while a boat requires 1 hour to make and 2 hours to finish. The toy company knows it will not sell anymore than 35 planes per week. Further, given the number of workers, the company cannot spend anymore than 160 hours per week finishing toys and 120 hours per week making toys. The company wishes to maximize the profit it makes by choosing how much of each toy to produce.

We can represent the profit maximization problem of the company as a linear programming problem. Let x_1 be the number of planes the company will produce and let x_2 be the number of boats the company will produce. The profit for each plane is $\$10 - \$3 = \$7$ per plane and the profit for each boat is $\$8 - \$2 = \$6$ per boat. Thus the total profit the company will make is:

$$z(x_1, x_2) = 7x_1 + 6x_2 \quad (1.1)$$

The company can spend no more than 120 hours per week making toys and since a plane takes 3 hours to make and a boat takes 1 hour to make we have:

$$3x_1 + x_2 \leq 120 \quad (1.2)$$

Likewise, the company can spend no more than 160 hours per week finishing toys and since it takes 1 hour to finish a plane and 2 hour to finish a boat we have:

$$x_1 + 2x_2 \leq 160 \quad (1.3)$$

Finally, we know that $x_1 \leq 35$, since the company will make no more than 35 planes per week. Thus the complete linear programming problem is given as:

$$\begin{aligned} \max \quad & z(x_1, x_2) = 7x_1 + 6x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \leq 120 \\ & x_1 + 2x_2 \leq 160 \\ & x_1 \leq 35 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned} \quad (1.4)$$

Exercise 1.1: Chemical Manufacturing

A chemical manufacturer produces three chemicals: A, B and C. These chemical are produced by two processes: 1 and 2.

- Running process 1 for 1 hour costs \$4 and yields 3 units of chemical A, 1 unit of chemical B and 1 unit of chemical C.
- Running process 2 for 1 hour costs \$1 and produces 1 units of chemical A, and 1 unit of chemical B (but none of Chemical C).

To meet customer demand,

- at least 10 units of chemical A,
- at least 5 units of chemical B and
- at least 3 units of chemical C

must be produced daily.

Assume that the chemical manufacturer wants to minimize the cost of production.

Develop a linear programming problem describing the constraints and objectives of the chemical manufacturer.

[Hint: Let x_1 be the amount of time Process 1 is executed and let x_2 be amount of time Process 2 is executed. Use the coefficients above to express the cost of running Process 1 for x_1 time and Process 2 for x_2 time. Do the same to compute the amount of chemicals A, B, and C that are produced.]

1.2 Modeling and Assumptions in Linear Programming

Outcomes

1. Address crucial assumptions when chosing to model a problem with linear programming.

A Generic Linear Program (LP)

Linear programming (LP) is a mathematical optimization framework widely used in decision-making and resource allocation. In its most basic form, an LP seeks to optimize (maximize or minimize) a linear objective function subject to a set of linear constraints. These constraints typically represent limitations or requirements in real-world scenarios, such as budgets, capacities, or resource availability.

Decision Variables:

The decision variables represent the quantities we want to determine. For example:

- x_i : Continuous variables ($x_i \in \mathbb{R}$, i.e., a real number), $\forall i = 1, \dots, n$.

These variables could represent amounts of products to produce, resources to allocate, or investments to make. We can use any letters to represent these variables. Typically we use x_i , but frequently will use y_i or other letters depending on the context and what makes reading the problem most convenient.

Parameters (Known Input Parameters):

Parameters are the given data in the problem, which are assumed to be known in advance:

- c_i : Cost coefficients for each decision variable, $\forall i = 1, \dots, n$. These values represent the contribution of each variable to the objective function.
- a_{ij} : Constraint coefficients, $\forall i = 1, \dots, n$ and $j = 1, \dots, m$. These define the relationships between decision variables and the constraints.
- b_j : Right-hand side values of the constraints, $\forall j = 1, \dots, m$. These represent limits or requirements for each constraint.

The general form of a linear program can be written as follows:

$$\begin{aligned}
 \max \quad & z = c_1x_1 + \cdots + c_nx_n \\
 \text{s.t.} \quad & a_{11}x_1 + \cdots + a_{1n}x_n \leq b_1 \\
 & \vdots \\
 & a_{m1}x_1 + \cdots + a_{mn}x_n \leq b_m \\
 & x_i \geq 0 \quad \forall i = 1, \dots, n.
 \end{aligned} \tag{1.1}$$

Features of a Linear Program

- **Objective Function:** The linear function $z = c_1x_1 + \cdots + c_nx_n$ represents the goal to optimize. This could involve maximizing profit, minimizing cost, or achieving some other measurable outcome.
- **Constraints:** The system of inequalities (or equalities) restricts the values of the decision variables. Each constraint reflects a real-world limitation, such as resource availability or capacity limits.
- **Feasibility:** The set of all possible solutions that satisfy the constraints is called the feasible region. The optimal solution must lie within this region.
- **Linear Relationships:** Both the objective function and constraints are linear, meaning the relationships between variables are additive and proportional.
- **Sets:** For specific optimization problem, it can be useful to define sets of variables or parameters to help write down the problem in a generic formulation.

Example of a Linear Program

To illustrate, consider a specific example with three decision variables and four constraints. This includes a mix of constraint types ($\leq$, $\geq$, and $=$), showcasing the versatility of linear programming:

Decision Variables:

x_i : Continuous variables ($x_i \in \mathbb{R}$, i.e., a real number), $\forall i = 1, \dots, 3$.

Parameters (Known Input Parameters):

- c_i : Cost coefficients, $\forall i = 1, \dots, 3$.
- a_{ij} : Constraint coefficients, $\forall i = 1, \dots, 3$ and $j = 1, \dots, 4$.
- b_j : Right-hand side values for each constraint, $\forall j = 1, \dots, 4$.

The linear program can be written as:

$$\text{Minimize } z = c_1x_1 + c_2x_2 + c_3x_3 \quad (1.2)$$

$$\text{Subject to } a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \geq b_1 \quad (1.3)$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \leq b_2 \quad (1.4)$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \quad (1.5)$$

$$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 \geq b_4 \quad (1.6)$$

$$x_1 \geq 0, x_2 \leq 0, x_3 \text{ unrestricted.} \quad (1.7)$$

A key feature of linear programming is the use of linear functions, defined as follows:

Definition 1.2: Linear Function

A function $z : \mathbb{R}^n \rightarrow \mathbb{R}$ is linear if there are constants $c_1, \dots, c_n \in \mathbb{R}$ such that:

$$z(x_1, \dots, x_n) = c_1x_1 + \dots + c_nx_n. \quad (1.8)$$

Linear functions are characterized by their simplicity and additivity, making them well-suited for optimization problems where proportionality is assumed. This simplicity allows linear programs to be solved efficiently, even for large-scale problems.

1.2.1. Assumptions

Inspecting Example 1.1 (or the more general problem in Equation (1.1)) we can see there are several assumptions that must be satisfied when using a linear programming model. We enumerate these below:

Proportionality Assumption A problem can be phrased as a linear program only if the contribution to the objective function *and* the left-hand-side of each constraint by each decision variable ($x_1, \dots, x_n$) is proportional to the value of the decision variable.

Additivity Assumption A problem can be phrased as a linear programming problem only if the contribution to the objective function *and* the left-hand-side of each constraint by any decision variable x_i ($i = 1, \dots, n$) is completely independent of any other decision variable x_j ($j \neq i$) and additive.

Divisibility Assumption A problem can be phrased as a linear programming problem only if the quantities represented by each decision variable are infinitely divisible (i.e., fractional answers make sense).

Certainty Assumption A problem can be phrased as a linear programming problem only if the coefficients in the objective function and constraints are known with certainty.

The first two assumptions simply assert (in English) that both the objective function and functions on the left-hand-side of the (in)equalities in the constraints are linear functions of the variables $x_1, \dots, x_n$.

The third assumption asserts that a valid optimal answer could contain fractional values for decision variables. It's important to understand how this assumption comes into play—even in the toy making example. Many quantities can be divided into non-integer values (ounces, pounds etc.) but many other quantities cannot be divided. For instance, can we really expect that it's reasonable to make $\frac{1}{2}$ a plane in the toy making example? When values must be constrained to true integer values, the linear programming problem is called an *integer programming problem*. There is a *vast* literature dealing with these problems. For many problems, particularly when the values of the decision variables may become large, a fractional optimal answer could be obtained and then rounded to the nearest integer to obtain a reasonable answer. For example, if our toy problem were re-written so that the optimal answer was to make 1045.3 planes, then we could round down to 1045.

The final assumption asserts that the coefficients (e.g., profit per plane or boat) is known with absolute certainty. In traditional linear programming, there is no lack of knowledge about the make up of the objective function, the coefficients in the left-hand-side of the constraints or the bounds on the right-hand-sides of the constraints. There is a literature on *stochastic programming* that relaxes some of these assumptions, but this too is outside the scope of the course.

Exercise 1.3

In a short sentence or two, discuss whether the problem given in Example 1.1 meets all of the assumptions of a scenario that can be modeled by a linear programming problem. Do the same for Exercise 1.1.

[Hint: Can you make $\frac{2}{3}$ of a toy? Can you run a process for $\frac{1}{3}$ of an hour?]

1.3 Examples

Outcomes

1. Identify decision variables and parameters
2. Learn how to format a linear optimization problem.
3. Identify and understand common classes of linear optimization problems.

We will begin with a few examples, and then discuss specific problem types that occur often.

Example 1.2: Production Planning

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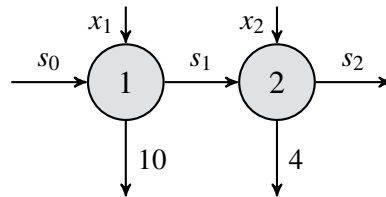
A manufacturing firm seeks to optimize its production schedule over a two-day planning horizon to meet daily demand while minimizing costs. The firm starts with an initial inventory of s_0 units. The costs associated with production and inventory are as follows:

- **Production costs:** \$10 per unit on Day 1, and \$16 per unit on Day 2.
- **Inventory cost:** \$5 per unit for carrying excess inventory from one day to the next.

Production

Inventory

Demand



The goal is to determine the number of units to produce each day and the inventory levels to minimize total costs while satisfying the demand for each day.

Solution. Sets:

- Days of production = $\{1, 2\}$.

Parameters:

- c_{p1} : Per unit production cost on Day 1 (\$10).
- c_{p2} : Per unit production cost on Day 2 (\$16).
- c_{inv} : Per unit inventory cost (\$5).
- d_1 : Demand on Day 1 (10 units).
- d_2 : Demand on Day 2 (4 units).
- s_0 : Inventory at the start of Day 1.

Decision variables:

- x_1 : Number of units produced on Day 1.
- x_2 : Number of units produced on Day 2.
- s_1 : Inventory at the end of Day 1.
- s_2 : Inventory at the end of Day 2.

Objective and Constraints:

$$\begin{array}{ll}
 \min & z = 10x_1 + 16x_2 + 5s_1 + 5s_2 & \text{(Total cost)} \\
 \text{s.t.} & s_0 + x_1 = 10 + s_1 & \text{(Day 1 balance)} \\
 & s_1 + x_2 = 4 + s_2 & \text{(Day 2 balance)} \\
 & x_1, x_2, s_0, s_1, s_2 \geq 0 & \text{(Non-negativity constraints).}
 \end{array}$$

**Definition 1.4: Optimization model components**

When defining an optimization model, it is essential to identify and clearly articulate the following five components:

- **Sets:** What lists of data do you need to write down? These could include indices, categories, or groups relevant to the problem.
- **Parameters (Data):** What is the actual data for the problem? These are the fixed numerical values such as costs, capacities, or resource availabilities.
- **Decision Variables:** What decisions need to be made to provide a solution? These variables represent the quantities that can be controlled or adjusted, such as production levels or resource allocations.
- **Objective Function:** The expression that we want to maximize or minimize. This is the goal of the optimization, such as minimizing cost or maximizing profit.
- **Constraints:** Certain rules that those decision variables should satisfy. These include limitations or requirements, such as resource capacities or demand fulfillment.

Each of these components plays a crucial role in formulating a well-structured optimization model that accurately represents the problem and provides actionable solutions. The role of Sets will become more important as we get into more complicated optimization models.

Example 1.3: Bakery Production

Excel PuLP Gurobipy

A small bakery produces three products each day: cakes, cookies, and muffins. Each item requires a specific amount of ingredients and baking time, and each contributes to the bakery's profit. The bakery operates with daily limits on flour, sugar, and baking time due to ingredient supply and staffing constraints.

Each cake earns a profit of \$6.00, requires 500 grams of flour, 200 grams of sugar, and 60 minutes of baking time. Each cookie earns a profit of \$1.50, requires 100 grams of flour, 50 grams of sugar, and 10 minutes of baking time. Each muffin earns a profit of \$2.50, requires 200 grams of flour, 80 grams of sugar, and 20 minutes of baking time.

The bakery has at most 4,800 grams of flour, 2,060 grams of sugar, and 480 minutes of baking time available each day.

Determine how many cakes, cookies, and muffins the bakery should produce each day in order to maximize its total profit, while staying within the limits on flour, sugar, and baking time.

Linear Programming Model

$$\begin{aligned}
 &\text{Maximize} && 6x_1 + 1.5x_2 + 2.5x_3 \\
 &\text{subject to} && 500x_1 + 100x_2 + 200x_3 \leq 4800 \quad (\text{Flour}) \\
 &&& 200x_1 + 50x_2 + 80x_3 \leq 2060 \quad (\text{Sugar}) \\
 &&& 60x_1 + 10x_2 + 20x_3 \leq 480 \quad (\text{Baking time}) \\
 &&& x_1, x_2, x_3 \geq 0 \\
 &&& x_1, x_2, x_3 \in \mathbb{Z} \quad (\text{if integer production is required})
 \end{aligned}$$

Example 1.4: Production with welding robot

Excel PuLP Gurobipy

You have 21 units of transparent aluminum alloy (TAA), LazWeld1, a joining robot leased for 23 hours, and CrumCut1, a cutting robot leased for 17 hours of aluminum cutting. You also have production code for a bookcase, desk, and cabinet, along with commitments to buy any of these you can produce for \$18, \$16, and \$10 apiece, respectively. A bookcase requires 2 units of TAA, 3 hours of joining, and 1 hour of cutting, a desk requires 2 units of TAA, 2 hours of joining, and 2 hour of cutting, and a cabinet requires 1 unit of TAA, 2 hours of joining, and 1 hour of cutting.

Formulate an LP to maximize your revenue given your current resources.

Solution.

Sets:

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- The types of objects = { bookcase, desk, cabinet }.

Parameters:

- Purchase cost of each object
- Units of TAA needed for each object
- Hours of joining needed for each object
- Hours of cutting needed for each object
- Hours of TAA, Joining, and Cutting available on robots

Decision variables:

x_i : number of units of product i to produce,
for all i =bookcase, desk, cabinet.

Objective and Constraints:

$$\begin{aligned}
 \max \quad & z = 18x_1 + 16x_2 + 10x_3 && \text{(profit)} \\
 \text{s.t.} \quad & 2x_1 + 2x_2 + 1x_3 \leq 21 && \text{(TAA)} \\
 & 3x_1 + 2x_2 + 2x_3 \leq 23 && \text{(LazWeld1)} \\
 & 1x_1 + 2x_2 + 1x_3 \leq 17 && \text{(CrumCut1)} \\
 & x_1, x_2, x_3 \geq 0.
 \end{aligned}$$



Example 1.5: The Diet Problem

Excel PuLP Gurobipy

In the future (as envisioned in a bad 70's science fiction film) all food is in tablet form, and there are four types, green, blue, yellow, and red. A balanced, futuristic diet requires, at least 20 units of Iron, 25 units of Vitamin B, 30 units of Vitamin C, and 15 units of Vitamin D. Formulate an LP that ensures a balanced diet at the minimum possible cost.

Tablet	Iron	B	C	D	Cost (\$)
Chem 1	6	6	7	4	1.25
Chem 2	4	5	4	9	1.05
Chem 3	5	2	5	6	0.85
Chem 4	3	6	3	2	0.65

Solution. Sets:

- Set of tablets {1,2,3,4}

Parameters:

- Iron in each tablet
- Vitamin B in each tablet
- Vitamin C in each tablet
- Vitamin D in each tablet
- Cost of each tablet

Decision variables:

x_i : number of tablet of type i to include in the diet, $\forall i \in \{1, 2, 3, 4\}$.

Objective and Constraints:

$$\begin{array}{ll}
 \text{Min} & z = 1.25x_1 + 1.05x_2 + 0.85x_3 + 0.65x_4 \\
 \text{s.t.} & 6x_1 + 4x_2 + 5x_3 + 3x_4 \geq 20 \\
 & 6x_1 + 5x_2 + 2x_3 + 6x_4 \geq 25 \\
 & 7x_1 + 4x_2 + 5x_3 + 3x_4 \geq 30 \\
 & 4x_1 + 9x_2 + 6x_3 + 2x_4 \geq 15 \\
 & x_1, x_2, x_3, x_4 \geq 0.
 \end{array}$$

**Example 1.6: The Next Diet Problem**

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Progress is important, and our last problem had too many tablets, so we are going to produce a single, purple, 10 gram tablet for our futuristic diet requires, which are at least 20 units of Iron, 25 units of Vitamin B, 30 units of Vitamin C, and 15 units of Vitamin D, and 2000 calories. The tablet is made from blending 4 nutritious chemicals; the following table shows the units of our nutrients per, and cost of, grams of each chemical.

Formulate an LP that ensures a balanced diet at the minimum possible cost.

Tablet	Iron	B	C	D	Calories	Cost (\$)
Chem 1	6	6	7	4	1000	1.25
Chem 2	4	5	4	9	250	1.05
Chem 3	5	2	5	6	850	0.85
Chem 4	3	6	3	2	750	0.65

Solution.Sets:

- Set of chemicals $\{1, 2, 3, 4\}$

Parameters:

- Iron in each chemical
- Vitamin B in each chemical
- Vitamin C in each chemical
- Vitamin D in each chemical
- Cost of each chemical

Decision variables:

x_i : grams of chemical i to include in the purple tablet, $\forall i = 1, 2, 3, 4$.

Objective and Constraints:

$$\begin{aligned}
 \min \quad & z = 1.25x_1 + 1.05x_2 + 0.85x_3 + 0.65x_4 \\
 \text{s.t.} \quad & 6x_1 + 4x_2 + 5x_3 + 3x_4 \geq 20 \\
 & 6x_1 + 5x_2 + 2x_3 + 6x_4 \geq 25 \\
 & 7x_1 + 4x_2 + 5x_3 + 3x_4 \geq 30 \\
 & 4x_1 + 9x_2 + 6x_3 + 2x_4 \geq 15 \\
 & 1000x_1 + 250x_2 + 850x_3 + 750x_4 \geq 2000 \\
 & x_1 + x_2 + x_3 + x_4 = 10 \\
 & x_1, x_2, x_3, x_4 \geq 0.
 \end{aligned}$$

**Example 1.7: Work Scheduling Problem**

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You are the manager of LP Burger. The following table shows the minimum number of employees required to staff the restaurant on each day of the week. Each employees must work for five consecutive days. Formulate an LP to find the minimum number of employees required to staff the restaurant.

Day of Week	Workers Required
1 = Monday	6
2 = Tuesday	4
3 = Wednesday	5
4 = Thursday	4
5 = Friday	3
6 = Saturday	7
7 = Sunday	7

Solution. This problem has multiple optimal solutions for which days workers begin working on, all of which result in 8 total workers hired.

Decision variables:

x_i : the number of workers that start 5 consecutive days of work on day i , $i = 1, \dots, 7$

Objective and Constraints:

$$\begin{array}{ll}
 \text{Min} & z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 \\
 \text{s.t.} & x_1 + x_4 + x_5 + x_6 + x_7 \geq 6 \\
 & x_2 + x_5 + x_6 + x_7 + x_1 \geq 4 \\
 & x_3 + x_6 + x_7 + x_1 + x_2 \geq 5 \\
 & x_4 + x_7 + x_1 + x_2 + x_3 \geq 4 \\
 & x_5 + x_1 + x_2 + x_3 + x_4 \geq 3 \\
 & x_6 + x_2 + x_3 + x_4 + x_5 \geq 7 \\
 & x_7 + x_3 + x_4 + x_5 + x_6 \geq 7 \\
 & x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0.
 \end{array}$$

One solution is as follows:

LP Solution	IP Solution
$z_{LP} = 7.333$	$z_I = 8.0$
$x_1 = 0$	$x_1 = 0$
$x_2 = 0.333$	$x_2 = 0$
$x_3 = 1$	$x_3 = 0$
$x_4 = 2.333$	$x_4 = 3$
$x_5 = 0$	$x_5 = 0$
$x_6 = 3.333$	$x_6 = 4$
$x_7 = 0.333$	$x_7 = 1$



Example 1.8: LP Burger - extended

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LP Burger has changed its policy, and allows, at most, two part time workers, who work for two consecutive days in a week. Formulate this problem.

Solution. Decision variables:

x_i : the number of workers that start 5 consecutive days of work on day i , $i = 1, \dots, 7$

y_i : the number of workers that start 2 consecutive days of work on day i , $i = 1, \dots, 7$.

Objective and Constraints:

$$\begin{aligned}
 &\text{Min} && z = 5(x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7) \\
 &&& + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7) \\
 &\text{s.t.} && x_1 + x_4 + x_5 + x_6 + x_7 + y_1 + y_7 \geq 6 \\
 &&& x_2 + x_5 + x_6 + x_7 + x_1 + y_2 + y_1 \geq 4 \\
 &&& x_3 + x_6 + x_7 + x_1 + x_2 + y_3 + y_2 \geq 5 \\
 &&& x_4 + x_7 + x_1 + x_2 + x_3 + y_4 + y_3 \geq 4 \\
 &&& x_5 + x_1 + x_2 + x_3 + x_4 + y_5 + y_4 \geq 3 \\
 &&& x_6 + x_2 + x_3 + x_4 + x_5 + y_6 + y_5 \geq 7 \\
 &&& x_7 + x_3 + x_4 + x_5 + x_6 + y_7 + y_6 \geq 7 \\
 &&& y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 \leq 2 \\
 &&& x_i \geq 0, y_i \geq 0, \forall i = 1, \dots, 7.
 \end{aligned}$$



1.3.1. Knapsack Problem

We consider now a simple example where the variables can only take the values of 0 or 1. We call these *binary variables*.

Example 1.9: Camping Trip

Excel PuLP Gurobipy

Imagine you are preparing for a week-long camping trip to the mountains. You have a backpack with a weight capacity of 20 kilograms. Your goal is to pack the most valuable items to ensure a comfortable and safe trip without exceeding the backpack's weight limit. Each item has a weight and a value associated with it, representing its importance and utility for the trip. The table below lists the potential items you can take, their weights in kilograms, and their values on a scale from 1 to 10 (with 10 being the most valuable).

Formulate an Integer Program to find the optimal items you should bring on your trip with you.

Item	Index	Weight (kg)	Value
Tent	A	4	10
Stove	B	2	9
Sleeping Bag	C	3	8
First Aid Kit	D	1	7
Food Supplies	E	5	9
Water Filter	F	1	6
Clothes	G	4	5
Map and Compass	H	0.5	7

Solution. Decision variables:

x_i : whether to include item i in the backpack, where $i \in \{A, B, C, D, E, F, G, H\}$, $x_i = 1$ if item i is included, and $x_i = 0$ otherwise.

Objective and Constraints:

$$\begin{aligned}
 \text{Max } z &= 10x_A + 9x_B + 8x_C + 7x_D + 9x_E + 6x_F + 5x_G + 7x_H \\
 \text{s.t. } &4x_A + 2x_B + 3x_C + 1x_D + 5x_E + 1x_F + 4x_G + 0.5x_H \leq 20 \\
 &x_A, x_B, x_C, x_D, x_E, x_F, x_G, x_H \in \{0, 1\}.
 \end{aligned}$$



1.3.2. Capital Investment

We next see a similar application of binary variables.

Example 1.10: Capital Allocation Problem

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You are a financial planner for an investment firm. The firm has \$100,000 to invest in a portfolio of projects. Each project has a projected return and requires an initial investment. The goal is to maximize the total return of the portfolio while not exceeding the available capital. The following table lists potential projects, their required investments, and their projected returns.

Formulate an LP to maximize the total return of the portfolio while not exceeding the available investment capital.

Project	Investment Required (\$)	Projected Return (\$)
A	20,000	30,000
B	30,000	40,000
C	50,000	75,000
D	10,000	15,000
E	40,000	60,000

Solution.

Sets:

- Projects: $P = \{A, B, C, D, E\}$

Parameters:

- inv_i : investment required for project i
- ret_i : projected return from project i
- $B = 100,000$: total available capital

Decision Variables:

$x_i \in \{0, 1\}$: 1 if project i is selected, 0 otherwise

Objective and Constraints:

$$\begin{aligned}
 \max \quad & z = 30,000x_A + 40,000x_B + 75,000x_C + 15,000x_D + 60,000x_E \\
 \text{s.t.} \quad & 20,000x_A + 30,000x_B + 50,000x_C + 10,000x_D + 40,000x_E \leq 100,000 \\
 & x_A, x_B, x_C, x_D, x_E \in \{0, 1\}
 \end{aligned}$$

Compact Form:

$$\begin{aligned}
& \max \quad \sum_{i \in P} \text{ret}_i \cdot x_i \\
& \text{s.t.} \quad \sum_{i \in P} \text{inv}_i \cdot x_i \leq B \\
& \quad x_i \in \{0, 1\} \quad \forall i \in P
\end{aligned}$$



1.3.3. Transportation problem

Example 1.11: Solar Panel Shipping

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SolTranz Inc. manufactures high-efficiency solar panels at three factories located in Reno (R), Boulder (B), and Tucson (T). The maximum production at each facility is 4,500, 5,500, and 7,000 panels, respectively. These panels must be shipped to five warehouses (labeled 1 through 5), each with a fixed demand.

The cost of producing a solar panel depends on the factory, and the shipping costs to each warehouse are also known. However, no more than 2,000 panels can be shipped from a factory to a single warehouse. The table below summarizes the data.

Formulate a linear program that minimizes the total cost of production and shipping to satisfy all warehouse demands.

Factory	Prod. Cost (\$)	WH1	WH2	WH3	WH4	WH5
Reno (R)	150	60	70	75	65	50
Boulder (B)	140	80	55	65	70	60
Tucson (T)	130	85	60	55	50	65

Shipping cost from each factory to each warehouse (in \$ per panel).

Warehouse Demands:

WH1: 3000 WH2: 3500 WH3: 2500 WH4: 4000 WH5: 3000

Solution.

Sets:

- Factories: $F = \{R, B, T\}$
- Warehouses: $W = \{1, 2, 3, 4, 5\}$

Parameters:

- C_f : production cost per panel at factory f
- S_f : supply capacity of factory f
- D_w : demand at warehouse w
- $T_{f,w}$: shipping cost per panel from factory f to warehouse w
- $M = 2000$: max panels shipped from any factory to any warehouse

Decision Variables:

$x_{f,w}$: number of solar panels shipped from factory f to warehouse w

Objective and Constraints:

$$\begin{aligned}
 \min \quad & \sum_{f \in F} \sum_{w \in W} (C_f + T_{f,w}) \cdot x_{f,w} \\
 \text{s.t.} \quad & \sum_{w \in W} x_{R,w} \leq 4500 && \text{(Reno capacity)} \\
 & \sum_{w \in W} x_{B,w} \leq 5500 && \text{(Boulder capacity)} \\
 & \sum_{w \in W} x_{T,w} \leq 7000 && \text{(Tucson capacity)} \\
 & \sum_{f \in F} x_{f,1} = 3000 && \text{(WH1 demand)} \\
 & \sum_{f \in F} x_{f,2} = 3500 && \text{(WH2 demand)} \\
 & \sum_{f \in F} x_{f,3} = 2500 && \text{(WH3 demand)} \\
 & \sum_{f \in F} x_{f,4} = 4000 && \text{(WH4 demand)} \\
 & \sum_{f \in F} x_{f,5} = 3000 && \text{(WH5 demand)} \\
 & 0 \leq x_{f,w} \leq 2000 && \forall f \in F, \forall w \in W
 \end{aligned}$$



1.3.4. Assignment Problem

Consider the assignment of 3 employees to 3 tasks, where each employee has a cost to do each task. How to figure out the best assignment to the tasks?

Example 1.12: Hiring for tasks

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In this assignment problem, we need to hire three people (Person 1, Person 2, Person 3) to three tasks (Task 1, Task 2, Task 3). In the table below, we list the cost of hiring each person for each task, in dollars. Since each person has a different cost for each task, we must make an assignment to minimize

	Cost	Task 1	Task 2	Task 3
our total cost.	Person 1	40	47	80
	Person 2	72	36	58
	Person 3	24	61	71

Given the specific costs of assigning three people to three tasks, we can write out the mathematical model explicitly using the given numbers.

Solution.

Sets: Set of tasks and set of persons

Parameters: Cost c_{ij} for person i to complete task j .

Variables: Let $x_{ij} = 1$ if person i is assigned to task j and 0 otherwise.

Objective Function:

Minimize the total cost of assignments:

$$z = 40x_{11} + 47x_{12} + 80x_{13} + 72x_{21} + 36x_{22} + 58x_{23} + 24x_{31} + 61x_{32} + 71x_{33} \quad (1.1)$$

Subject to Constraints:

Each person is assigned to exactly one task:

$$x_{11} + x_{12} + x_{13} = 1 \quad (1.2)$$

$$x_{21} + x_{22} + x_{23} = 1 \quad (1.3)$$

$$x_{31} + x_{32} + x_{33} = 1 \quad (1.4)$$

Each task is assigned to exactly one person:

$$x_{11} + x_{21} + x_{31} = 1 \quad (1.5)$$

$$x_{12} + x_{22} + x_{32} = 1 \quad (1.6)$$

$$x_{13} + x_{23} + x_{33} = 1 \quad (1.7)$$

Binary constraints on the variables:

$$x_{ij} \in \{0, 1\} \quad \forall i \in \{1, 2, 3\}, \forall j \in \{1, 2, 3\} \quad (1.8)$$



1.4 Exercises

Exercise 1.13 Which ones are linear functions?

1. $x^2 + 3y + 2z$
2. $\sqrt{x_1 x_2 x_3}$
3. $\sum_{i=1}^n a_i 2^{x_i}$, where $a_1, a_2, \dots, a_n$ are constant
4. $\sum_{i=1}^n a_i x_i$, where $a_1, a_2, \dots, a_n$ are constant

Exercise 1.14 Which one is a linear function?

1. $2x_1 + x_2 x_3 - 7x_3$
2. $\sqrt{2}x_1 + x_2 + 3^{1.6}x_3$

Exercise 1.15 Which ones are linear inequalities? Can any be made into linear inequalities?

1. $\frac{x_1}{x_1 + x_2 + x_3} \geq 0.5$
2. $\sin x_1 + x_2 + x_3 = 0.5$
3. $\sum_{i=1}^n a_i 2^{x_i} \geq b$, where $a_1, a_2, \dots, a_n, b$ are constant
4. $\sum_{i=1}^n a_i x_i = b$, where $a_1, a_2, \dots, a_n, b$ are constant
5. $\sum_{i=1}^n a_i x_i \geq b$, where $a_1, a_2, \dots, a_n, b$ are constant

Exercise 1.16 An appliance store sells three types of ovens: small, medium, and large. The small, medium, and large ovens require, respectively, 3, 5, and 6 cubic feet of storage space; at most 100 cubic feet of storage is available. Each oven takes 1 hour of sales time; there is a maximum of 200 hours of sales labor time available. The small, medium, and large ovens require, respectively, 1, 1, and 2 hours of installation time; at most 280 hours of installer labor are available monthly.

If the profit (in dollars) from small, medium, and large ovens is 50, 100, and 150 respectively, how many of each type of oven should be sold to maximize profit, and what is the maximum profit?

Exercise 1.17 A factory manufactures three products, A, B, and C. Each product requires the use of two machines (Machine I and Machine II). The total hours available per month on Machine I and Machine II

are 180 and 300, respectively. The table below shows the time requirements (in hours per unit) and profit per unit for each product.

	A	B	C
Machine I	1	2	2
Machine II	2	2	4
Profit (dollars)	20	30	40

How many units of each product should be manufactured to maximize profit, and what is the maximum profit?

Exercise 1.18 A company produces three products, A, B, and C, at two factories, Factory I and Factory II. Daily production of each factory for each product is listed below:

	Factory I	Factory II
Product A	10	20
Product B	20	20
Product C	20	10

The company must produce at least 1000 units of product A, 1600 units of product B, and 700 units of product C. If the cost of operating Factory I is \$4000 per day and the cost of operating Factory II is \$5000 per day, how many days should each factory operate to complete the order at a minimum cost, and what is the minimum cost?

Exercise 1.19 Professor Wright gives three types of quizzes (objective, recall, and recall-plus) and decides to give at least 20 quizzes next quarter. Each type of quiz requires the students to spend extra preparation time:

Objective: 10 minutes per quiz,

Recall: 30 minutes per quiz,

Recall-plus: 60 minutes per quiz.

He wants students to spend at least 12 hours (720 minutes) in total preparation for these quizzes (beyond normal study time).

Average quiz scores are:

Objective: 5, Recall: 6, Recall-plus: 7.

He wants the total of all quiz scores to be at least 130 points.

It takes the professor:

1 minute to grade each objective quiz,

2 minutes to grade each recall quiz,

3 minutes to grade each recall-plus quiz.

How many of each type should he give in order to minimize his own grading time?

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